團體程式作業

分工表：

內容	學號	姓名
Chap2：建好環境框架如 module_init、module_exit、/proc 介面	112820018	許景喬
Chap2：實作 /proc/seconds	112820042	杜詠霖
Chap3 Project1：建好環境框架如主迴圈 + fork/exec	112820025	陳翊揚
Chap3 Project1：實作 Redirection + Pipe	112820003	辛政隆

- chap2
  - 編譯方式：在 Maefile 所在位置 $make
  - 執行方式：
    - $make 編譯組
    - $sudo insmod jiffies.ko 和 $sudo insmod seconds.ko 載入模組
    - $cat /proc/jiffies 讀取 /proc/jiffies 的檔案內容(輸出數字，重複執行數字會變大)
    - $sudo dmesg | tail -n 1 取得最後一個系統日誌
    - $sudo rmmod jiffies.ko 刪除模組
    - $cat /proc/seconds 首次載入模組後經過的秒數
    - $./test_modules.sh 進行測試
  - 執行結果：

```
vboxuser@Ubuntu22:~/Desktop/os/114-2_Operator_System_Team_Project/HW1/Chap2$ make
make -C /lib/modules/6.8.0-106-generic/build M=/home/vboxuser/Desktop/os/114-2_Operator_System_Team_Project/HW1/Chap2 modules
make[1]: Entering directory '/usr/src/linux-headers-6.8.0-106-generic'
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04.3) 12.3.0
  You are using:           gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04.3) 12.3.0
make[1]: Leaving directory '/usr/src/linux-headers-6.8.0-106-generic'
vboxuser@Ubuntu22:~/Desktop/os/114-2_Operator_System_Team_Project/HW1/Chap2$ sudo insmod jiffies.ko
vboxuser@Ubuntu22:~/Desktop/os/114-2_Operator_System_Team_Project/HW1/Chap2$ cat /proc/jiffies
4320401085
vboxuser@Ubuntu22:~/Desktop/os/114-2_Operator_System_Team_Project/HW1/Chap2$ cat /proc/jiffies
4320403150
vboxuser@Ubuntu22:~/Desktop/os/114-2_Operator_System_Team_Project/HW1/Chap2$ cat /proc/jiffies
4320404498
vboxuser@Ubuntu22:~/Desktop/os/114-2_Operator_System_Team_Project/HW1/Chap2$ sudo dmesg | tail -n 1
[25729.393925] /proc/jiffies created
vboxuser@Ubuntu22:~/Desktop/os/114-2_Operator_System_Team_Project/HW1/Chap2$ sudo rmmod jiffies
vboxuser@Ubuntu22:~/Desktop/os/114-2_Operator_System_Team_Project/HW1/Chap2$ sudo dmesg | tail -n 2
[25729.393925] /proc/jiffies created
[25765.498105] /proc/jiffies removed
vboxuser@Ubuntu22:~/Desktop/os/114-2_Operator_System_Team_Project/HW1/Chap2$ cat /proc/jiffies
cat: /proc/jiffies: No such file or directory
vboxuser@Ubuntu22:~/Desktop/os/114-2_Operator_System_Team_Project/HW1/Chap2$
```

```
br@br:~/桌面/114-2_Operator_System_Team_Project/HW1/Chap2$ make
make -C /lib/modules/6.8.0-106-generic/build M=/home/br/桌面/114-2_Operator_System_Team_Project/HW1/Chap2 m
odules
make[1]: 進入目錄「/usr/src/linux-headers-6.8.0-106-generic」
warning: the compiler differs from the one used to build the kernel
  The kernel was built by: x86_64-linux-gnu-gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04.3) 12.3.0
  You are using:           gcc-12 (Ubuntu 12.3.0-1ubuntu1~22.04.3) 12.3.0
make[1]: 離開目錄「/usr/src/linux-headers-6.8.0-106-generic」
br@br:~/桌面/114-2_Operator_System_Team_Project/HW1/Chap2$ sudo iamod seconds.ko
sudo: iamod：找不到指令
br@br:~/桌面/114-2_Operator_System_Team_Project/HW1/Chap2$ sudo ismod seconds.ko
sudo: ismod：找不到指令
br@br:~/桌面/114-2_Operator_System_Team_Project/HW1/Chap2$ sudo insmod seconds.ko
br@br:~/桌面/114-2_Operator_System_Team_Project/HW1/Chap2$ ls /proc/seconds
/proc/seconds
br@br:~/桌面/114-2_Operator_System_Team_Project/HW1/Chap2$ cat /proc/seconds
47
br@br:~/桌面/114-2_Operator_System_Team_Project/HW1/Chap2$ cat /proc/seconds
52
br@br:~/桌面/114-2_Operator_System_Team_Project/HW1/Chap2$ sudo remod seconds.ko
sudo: remod：找不到指令
br@br:~/桌面/114-2_Operator_System_Team_Project/HW1/Chap2$ sudo rmmod seconds.ko
br@br:~/桌面/114-2_Operator_System_Team_Project/HW1/Chap2$ ls /proc/seconds
ls: 無法存取 '/proc/seconds': 沒有此一檔案或目錄
br@br:~/桌面/114-2_Operator_System_Team_Project/HW1/Chap2$
```

- Chap3 Project1
  - 編譯方式：$ g++ code.cpp -o code
  - 執行方式：
    - $./code 進入 Unix shell
    - ls 列出當前目錄的所有檔案與子目錄
    - ls -a 列出當前目錄下的所有檔案與子目錄以及隱藏檔
    - ls -la 列出當前目錄下的所有檔案與子目錄以及隱藏檔和他們的權限
    - history 列出歷史紀錄
    - !! 前一個指令
    - echo hello > test.txt 將 hello 導入至 test.txt 裡
    - cat < test.txt 將檔案內容給 cat 處理
    - ls -la | cat 將 ls -la 得到的內容傳給 cat
    - exit 離開
  - 執行結果：

```
ianchen@IanChen:/mnt/d/北科大/114-2作業系統/114-2_Operator_System_Team_Project/HW1/Chap3$ g++ ./code.cpp -o code
ianchen@IanChen:/mnt/d/北科大/114-2作業系統/114-2_Operator_System_Team_Project/HW1/Chap3$ ./code
osh>ls
code  code.cpp
osh>ls -a
.  ..  code  code.cpp
osh>ls -la
total 60
drwxrwxrwx 1 ianchen ianchen  4096 Mar 28 20:54 .
drwxrwxrwx 1 ianchen ianchen  4096 Mar 28 15:40 ..
-rwxrwxrwx 1 ianchen ianchen 50968 Mar 28 20:54 code
-rwxrwxrwx 1 ianchen ianchen  5800 Mar 28 20:54 code.cpp
osh>history
1 ls
2 ls -a
3 ls -la
4 history
osh>!!
history
1 ls
2 ls -a
3 ls -la
4 history
5 history
osh>sleep 5 &
[背景執行] PID = 97826
osh>sleep 5
osh>exit
```

```
creepertail@creepertail:/mnt/c/Users/pogcr/Desktop/ntut/operating_system/hw1/OSHW#1_teambased_112820003_112820018_112820025_112820042/Chap3$ ./code
osh>echo hello > test.txt
osh>cat < test.txt
hello
osh>ls -la | cat
total 60
drwxrwxrwx 1 creepertail creepertail  4096 Mar 30 22:25 .
drwxrwxrwx 1 creepertail creepertail  4096 Mar 30 21:57 ..
-rwxrwxrwx 1 creepertail creepertail 52576 Mar 30 22:10 code
-rwxrwxrwx 1 creepertail creepertail  8172 Mar 29 21:52 code.cpp
osh>
```